

Product Requirement Document (PRD)

AI-Powered Customer Support Ticket Triage - SaaS Delivery Simulation

Prepared by

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Document Type	Product Requirement Document
Project Nature	Independent SaaS and AI delivery simulation
Product Domain	Customer experience operations and support workflow automation
Primary Output	Ticket classification, priority assignment, routing recommendation, and manual review flag
Intended Audience	Product, implementation, technology, support, and customer success teams

Confidentiality and usage note: This document is prepared as a professional portfolio artifact for a simulated SaaS implementation. It does not represent a live client deployment and does not include real customer data.

1. Executive Summary

This Product Requirement Document defines the functional and non-functional requirements for an AI-enabled SaaS workflow that supports customer support ticket triage. The workflow is designed for a mock e-commerce client, UrbanCart, where support tickets are currently interpreted and sorted manually by support agents. The proposed product capability classifies each ticket into a predefined category, assigns an operational priority, recommends the correct routing team, and flags unclear tickets for manual review.

The purpose of this PRD is to translate business needs into implementation-ready product requirements. It connects customer experience objectives with platform behavior, user roles, workflow rules, acceptance criteria, data expectations, and quality controls. The document is intentionally structured to support collaboration between product, implementation, technology, support, and customer success teams.

The solution is a simulation of AI-assisted ticket triage. Classification behavior is represented through explainable rules and workflow logic rather than a trained production machine learning model. This keeps the project transparent, auditable, and appropriate for a portfolio-based SaaS delivery artifact.

2. Product Objective and Expected Outcomes

The product objective is to support faster, more consistent, and more transparent handling of customer support tickets by introducing an AI-enabled triage layer inside a SaaS customer support workflow.

Expected outcomes include:

- Reduce manual effort required to read, categorize, prioritize, and route incoming support tickets.
- Improve consistency of support operations by applying standardized category, priority, and routing rules.
- Increase visibility for support managers by making ticket urgency and ownership clearer.
- Support implementation teams with a documented requirement baseline that can be reviewed, demoed, and iterated.
- Provide a practical example of how AI-enabled workflow automation can be scoped, documented, and delivered inside a SaaS product environment.

3. Product Context

UrbanCart is a simulated e-commerce business that receives customer support requests across common issue areas such as refunds, delayed deliveries, failed payments, cancellation requests, damaged products, login issues, and general order queries. Without a structured triage process, support teams may experience inconsistent routing, slower first response, and difficulty identifying urgent cases.

The proposed SaaS workflow introduces a classification and routing layer between ticket intake and agent handling. This layer does not replace support agents. Instead, it improves the quality and speed of the first operational decision by suggesting category, priority, routing team, and manual review status.

4. Target Users and Personas

User / Persona	Primary Need	Product Interaction
Support Agent	Handle tickets efficiently with less manual sorting.	Views category, priority, routing team, and manual review status before responding.
Support Manager	Monitor ticket urgency, workload, and escalation patterns.	Filters high-priority tickets, reviews manual review cases, and monitors category distribution.
Implementation Team Member	Translate client requirements into workflow configuration.	Uses PRD, BRD, user stories, and routing rules to configure and validate the workflow.
Product Team Member	Define product behavior and manage scope.	Reviews requirements, acceptance criteria, edge cases, and release readiness.
Technology Team Member	Build or configure classification and routing logic.	Implements data fields, rules, validation checks, and integration behavior.

User / Persona	Primary Need	Product Interaction
Customer Success Team Member	Support client onboarding and adoption.	Uses demo script and training guide to explain workflow behavior to business users.

5. Product Scope

5.1 In Scope

- Ticket intake from a text-based customer message.
- Classification of ticket category using predefined categories.
- Priority assignment using defined urgency and issue-type rules.
- Routing recommendation to the correct support team.
- Manual review flag when the system cannot confidently categorize a ticket.
- User-visible output fields for category, priority, routing team, and review status.
- Sample data and documentation to demonstrate the workflow.
- Acceptance criteria, quality gates, and implementation notes for product delivery review.

5.2 Out of Scope

- Training or deployment of a production AI or machine learning model.
- Processing of real customer data or personally identifiable information.
- Live integration with payment, logistics, CRM, email, or chat systems.
- Production-grade authentication, authorization, billing, and tenant management.
- Fully automated resolution of customer issues without human agent review.
- Enterprise analytics beyond the simulated reporting metrics described in this PRD.

6. Feature Overview

Feature Area	Description	Business Value
Ticket Classification	Identify the issue type from the customer message and assign a standardized category.	Improves consistency and reduces agent sorting effort.
Priority Assignment	Assign High, Medium, or Low priority based on urgency, issue type, and escalation indicators.	Helps teams respond to urgent issues faster.
Routing Recommendation	Map ticket category to the responsible internal support team.	Improves ownership clarity and reduces misrouting.
Manual Review Flag	Flag unclear or unmatched tickets for human validation.	Reduces risk of incorrect classification and routing.
Agent Visibility	Display ticket ID, message, category, priority, routing team, and review status.	Supports faster decision-making by support agents.
Manager Monitoring	Enable visibility into high-priority and manual-review tickets.	Supports operational oversight and continuous improvement.

7. Functional Requirements

ID	Requirement	Acceptance Criteria	Priority
FR-001	Classify incoming tickets into predefined categories.	Each ticket receives one category from the approved ticket taxonomy or is marked for manual review.	High
FR-002	Assign ticket priority as High, Medium, or Low.	Priority is generated for every processed ticket and displayed to the support user.	High
FR-003	Recommend routing team based on assigned category.	Each supported category maps to one defined routing team.	High

ID	Requirement	Acceptance Criteria	Priority
FR-004	Flag unclear tickets for manual review.	Tickets that do not match defined categories are marked Manual Review and routed to General Support.	High
FR-005	Display ticket triage output clearly to support agents.	Ticket ID, customer message, category, priority, routing team, and review flag are visible.	Medium
FR-006	Support high-priority visibility for managers.	High-priority tickets can be identified using the priority field.	Medium
FR-007	Maintain explainable classification logic.	Rules or reasoning used for classification can be described in documentation.	Medium
FR-008	Allow category taxonomy to be reviewed before implementation.	All categories are documented and approved before workflow validation.	Medium
FR-009	Support exception handling for unsupported ticket types.	Unsupported or ambiguous tickets are not forced into incorrect categories.	High
FR-010	Provide sample data for validation.	Sample tickets cover all supported categories and priority levels.	Medium
FR-011	Support demo-ready workflow explanation.	The feature can be explained through a step-by-step demo script.	Medium
FR-012	Support training documentation for end users.	Support agents and managers can understand how to interpret output fields.	Medium

8. Non-Functional Requirements

ID	Category	Requirement	Rationale
NFR-001	Usability	The workflow output should be easy for support users to understand without technical knowledge.	Supports adoption by support agents and managers.
NFR-002	Explainability	Classification and priority logic should be documented and understandable.	Builds trust and supports review by implementation teams.
NFR-003	Data Privacy	The simulation should not use real customer data.	Avoids privacy risk in a portfolio project.
NFR-004	Maintainability	Categories, priority rules, and routing logic should be documented for future updates.	Allows iterative refinement as client needs change.
NFR-005	Reliability	The workflow should return a consistent output structure for each ticket.	Enables predictable downstream handling.
NFR-006	Auditability	Rules, assumptions, edge cases, and manual review criteria should be captured.	Supports traceability and governance.
NFR-007	Scalability Readiness	The design should allow categories and teams to be extended later.	Supports future SaaS configuration scenarios.
NFR-008	Documentation Quality	Product documentation should be clear for both business and technical stakeholders.	Supports implementation, demo, handover, and training.

9. Ticket Taxonomy and Routing Logic

The product uses a predefined ticket taxonomy to create consistent support operations. Each category is mapped to a routing team and a primary operational objective.

Category	Example Customer Intent	Routing Team	Operational Objective
Refund	Refund status, refund request, money return.	Billing and Refund Team	Resolve refund or billing-status issues.
Delivery Delay	Order not received, delayed delivery, late shipment.	Logistics Team	Investigate delivery timeline and logistics status.
Payment Issue	Payment failed, money deducted, order not placed.	Payments Team	Resolve payment confirmation or transaction issues.
Damaged Product	Broken, defective, damaged, replacement request.	Returns and Replacement Team	Handle replacement, return, or quality concern.

Category	Example Customer Intent	Routing Team	Operational Objective
Cancellation	Cancel order, order placed by mistake.	Order Management Team	Process or evaluate cancellation request.
Login Issue	Password reset, account access, login failure.	Technical Support Team	Resolve access and authentication support requests.
General Query	Address update, order information, basic question.	General Support Team	Handle informational or non-critical queries.

10. Priority and Escalation Model

Priority	Trigger Conditions	Expected Handling Logic
High	Payment failure with money deducted, repeated complaint, damaged product with urgent tone, escalation language, severe dissatisfaction.	Flag prominently and route to the responsible team for faster handling.
Medium	Refund pending, delivery delayed, cancellation request, non-critical complaint requiring team action.	Route to the correct team with normal operational visibility.
Low	General query, account help, order status request, address update, basic informational request.	Route to general or relevant support queue without escalation.
Manual Review	Ambiguous message, unsupported issue type, insufficient context, unclear intent.	Route to General Support for human validation before final handling.

Escalation behavior is intentionally conservative. The product should avoid over-automating sensitive decisions. Where uncertainty exists, manual review is preferred over incorrect routing.

11. User Flow and System Behavior

Step	Action	System Output
1	Customer submits a support ticket.	Ticket message is captured for triage.
2	Ticket enters the SaaS support workflow.	Ticket ID and message are available to the triage layer.
3	AI-enabled classification logic evaluates the ticket text.	Issue category is assigned or manual review is triggered.
4	Priority rules evaluate urgency and impact.	Priority level is assigned as High, Medium, or Low.
5	Routing rules map category to team.	Suggested routing team is generated.
6	Support agent reviews triage output.	Agent can use category, priority, and team recommendation.
7	Manager monitors operational view.	High-priority and manual review tickets can be tracked.

12. Data Inputs and Output Schema

The simulation uses a simple structured schema to represent ticket triage results. This schema can be used for sample CSV data, dashboard output, or API response documentation in a future implementation.

Field	Type	Description	Example
ticket_id	String	Unique identifier for the support ticket.	TCK-001
customer_message	Text	Original customer support message.	My payment was deducted but the order was not placed.
predicted_category	String	Assigned ticket category.	Payment Issue
priority	String	Operational priority level.	High
routing_team	String	Recommended team responsible for handling.	Payments Team
manual_review_flag	Boolean	Indicates whether human validation is required.	False
classification_reason	Text	Brief explainable reason for the category or priority.	Payment-related keywords and urgency detected.

13. Requirement Traceability Matrix

Business Need	Product Requirement	Validation Method
Reduce manual ticket sorting.	FR-001, FR-002, FR-003	Run sample tickets through category, priority, and routing rules.
Improve handling of urgent issues.	FR-002, FR-006	Verify High-priority outputs for payment failure and escalation cases.
Avoid incorrect routing when intent is unclear.	FR-004, FR-009	Test ambiguous tickets and confirm Manual Review behavior.
Support team adoption.	FR-005, FR-011, FR-012, NFR-001	Review demo script, training guide, and output visibility.
Maintain explainable workflow behavior.	FR-007, NFR-002, NFR-006	Review documented classification, assumptions, and edge cases.

14. Acceptance Criteria and Quality Gates

Quality Gate	Acceptance Standard
Category Coverage	Sample data includes all supported categories listed in the product taxonomy.
Priority Coverage	Sample data includes High, Medium, Low, and Manual Review scenarios.
Routing Accuracy	Each category maps to the expected support team.
Explainability	Classification and priority rules can be explained to a non-technical stakeholder.
Documentation Consistency	BRD, PRD, user stories, implementation plan, demo script, and training guide remain aligned.
Privacy Safety	No real customer data is used in the project.
Demo Readiness	A reviewer can understand the workflow using the demo script and sample tickets.

15. Product Metrics and Reporting Considerations

Although this is a simulation, a production version of the workflow could be evaluated through operational and adoption-oriented metrics. These metrics would help the implementation and support teams understand whether the product capability is improving customer experience operations.

Metric	Purpose
Ticket category distribution	Shows which issue areas create the highest support volume.
High-priority ticket count	Helps managers track urgent customer issues.
Manual review rate	Highlights where classification rules may need improvement.
Routing accuracy feedback	Shows whether tickets are reaching the correct teams.
Average first response readiness	Measures whether triage output helps agents respond faster.
Repeat issue patterns	Helps product and support teams identify recurring customer pain points.

16. Product Risks and Mitigations

Risk	Impact	Mitigation
Incorrect category assignment	Ticket may be routed to the wrong team.	Use manual review for ambiguous cases and document category rules clearly.
Over-prioritization of non-urgent tickets	Support queues may become noisy.	Define priority rules and review High-priority examples.
Under-prioritization of urgent tickets	Customer escalation may be delayed.	Include urgency keywords and repeated complaint indicators in logic.

Risk	Impact	Mitigation
Unclear client requirements	Implementation may drift from business expectations.	Maintain BRD, PRD, open questions, and traceability matrix.
Insufficient user adoption	Support teams may ignore triage output.	Provide demo script, training guide, and clear field definitions.
Privacy concern	Use of real customer information would create unnecessary risk.	Use simulated data only and state this clearly in documentation.

17. Open Questions for Future Iteration

- Should the client be able to configure custom ticket categories by business unit or geography?
- Should VIP customers or premium support tiers automatically receive elevated priority?
- Should unresolved High-priority tickets trigger an escalation workflow?
- Should support managers be able to override category, priority, or routing decisions?
- Should classification rules be reviewed periodically based on manual review trends?
- Should a future version include multilingual ticket classification?
- Should the workflow integrate with email, chat, CRM, or helpdesk systems?

18. Handover and Portfolio Positioning

This PRD can be handed over with the BRD, user stories, implementation plan, demo script, training guide, workflow diagram, and sample ticket dataset. Together, these artifacts demonstrate how a business problem can be translated into product requirements and delivery-ready documentation for a SaaS-based AI workflow.

For interview discussion, the project should be positioned as an independent SaaS delivery simulation that demonstrates requirement documentation, stakeholder thinking, feature scoping, acceptance criteria, workflow design, and AI-enabled customer support automation concepts. It should not be described as a live client implementation or a production AI model.